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Bihar Engineering University, Patna

B.Tech 3rd Semester Exam., 2024

Code: 100311	***Subject: MATHEMATICS-III***	***Time: 03 Hours***
Semester: III	***(DIFFERENTIAL CALCULUS)***	***Full Marks: 70***

Instructions:-

(i) *The marks are indicated in the right-hand margin.*
(ii) *There are NINE questions in this paper.*
(iii) *Attempt FIVE questions in all.*
(iv) *Question No. 1 is compulsory.*

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) The solution of the differential equation $y'' - 3y' + 2y = e^{3x}$

(i) $C_1e^x + C_2e^{2x} + e^{3x}/2$
(ii) $C_1e^{-x} + C_2e^{-2x} + e^{3x}/2$
(iii) $C_1e^{-x} + C_2e^{2x} + e^{3x}/2$
(iv) None of these

(b) The solution of $p + q = z$ is

(i) $f(xy, y \log z) = 0$
(ii) $f(x + y, y + \log z) = 0$
(iii) $f(x - y, y - \log z) = 0$
(iv) None of these

(c) The series $1 - (1/(1!)^2)(x/2)^2 + (1/(2!)^2)(x/2)^4 - (1/(3!)^2)(x/2)^6 + \ldots$ is

(i) $J_{1/2}(x)$
(ii) $xJ_{1/2}(x)$
(iii) $xJ_0(x)$
(iv) None of these

(d) If $\int_{-1}^{1} P_n(x)\,dx = 2$, then n is

(i) 0
(ii) 1
(iii) 2
(iv) None of these

(e) The particular solution of $y'' + 9y' = \sin 3x$ is

(i) $-(x/6)\cos 3x$
(ii) $(x/6)\cos 3x$
(iii) $-(3x/2)\cos 3x$
(iv) $(3x/2)\cos 3x$

(f) The differential equation $y'' + (s^3 \sin x)^5 y' + y = \cos x^3$ is

(i) Homogeneous
(ii) Non-linear
(iii) Second order linear
(iv) Non-homogeneous with constant coefficient

(g) The magnitude of the directional derivative of the function $f(x, y) = x^2 + 3y^2$ in a direction normal to the circle $x^2 + y^2 = 2$, at the point (1,1) is

(i) $4\sqrt{2}$
(ii) $5\sqrt{2}$
(iii) $7\sqrt{2}$
(iv) $9\sqrt{2}$

(h) If $R = xi + yj + zk$ is the position vector, then div R is

(i) 1
(ii) 3
(iii) −3
(iv) −2

(i) The degree of the homogeneous function $f(x, y) = (x^{7/3} + y^{7/3})/(x + y)$ is

(i) 4/5
(ii) 4/3
(iii) 7/3
(iv) None of these

(j) If $f(x, y) = \tan^{-1} y/x + \sin^{-1} x/y$, then $x\partial f/\partial x + y\partial f/\partial y =$

(i) 2
(ii) 1
(iii) 0
(iv) −1

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Q.2 **[7+7=14]**

(a) If $y^{1/m} + y^{-1/m} = 2x$, prove that
(i) $(x^2 - 1)y_2 + xy_1 - m^2y = 0$
(ii) $(x^2 - 1)y_{n+2} + (2n + 1)xy_{n+1} + (n^2 - m^2)y_n = 0$

(b) If $p^2 = a^2 \cos^2\theta + b^2 \sin^2\theta$, prove that $p + d^2p/d\theta^2 = a^2b^2/p^3$

Q.3 **[7+7=14]**

(a) Use Taylor's theorem to expand $f(x, y) = x^y$ in powers of $(x - 1)$ and $(y - 1)$.

(b) Find the dimension of the rectangular box, open at the top, of maximum capacity whose surface area is 432 sq.cm.

Q.4 **[7+7=14]**

(a) If $u = \tan^{-1}(y^2/x)$, prove that $x^2\,\partial^2u/\partial x^2 + 2xy\,\partial^2u/\partial x\partial y + y^2\,\partial^2u/\partial y^2 = -\sin^2u \cdot \sin 2u$

(b) If by the substitution $u = x^2 - y^2$, $v = 2xy$, $f(x, y) = \theta(u, v)$, show that $\partial^2f/\partial x^2 + \partial^2f/\partial y^2 = 4(x^2 + y^2)(\partial^2\theta/\partial u^2 + \partial^2\theta/\partial v^2)$

Q.5 **[7+7=14]**

(a) Verify Stoke's theorem for $F = (2x - y)i - yz^2j - y^2zk$ over the upper half surface $x^2 + y^2 + z^2 = 1$, bounded by its projection on the xy −plane.

(b) If the directional derivative of $\varphi = ax^2y + by^2z + cxz^2$ at the point (1,1,1) has maximum magnitude 15 in the direction parallel to the line $(x - 1)/2 = (y - 3)/(-2) = z$, find the values of a, b, c

Q.6 **[7+7=14]**

(a) Solve $(x^2 + y^2)(x\,dx + y\,dy) = a^2(x\,dy - y\,dx)$

(b) Solve $p(p + x) = y(x + y)$

Q.7 **[7+7=14]**

(a) Solve $(2y \log y + y - x)dy = y\,dx$

(b) Solve, by the method of variation of parameters, $y'' - 2y' + y = e^x \log x$.

Q.8 **[7+7=14]**

(a) Solve $x^2\,d^2y/dx^2 - 3x\,dy/dx + y = \log x \cdot (\sin(\log x) + 1)/x$

(b) Prove that $J_m(x) = (1/\pi)\int_0^\pi \cos(m\theta - x \sin\theta)\,d\theta$, where 'n' is positive integer

Q.9 **[7+7=14]**

(a) Solve the equation : $(z^2 - 2yz - y^2)p + (xy + zx)q = xy - zx$.

(b) Solve the equation : $(x - y)(px - qy) = (p - q)^2$

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